

1 Statistical Issues in Multicenter Randomized Clinical
2 Trials

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5 **Abstract**

6 Multicenter randomized controlled trials require a choice among ignor-
7 ing, fixing, or randomizing site effects in the primary analysis, and this
8 choice interacts with stratified-randomization adjustment, treatment-by-site
9 interaction, and site-size balance. We report a factorial Monte Carlo sim-
10 ulation, structured under the ADEMP framework of Morris, White, and
11 Crowther (2019), that jointly varies the number of sites, site-level variance,
12 treatment-by-site interaction, and site-size balance, comparing OLS ignoring
13 site, OLS with fixed site effects, and a linear mixed model with random site
14 effects. In addition to bias, empirical and model-based standard error, power,
15 and coverage under a true alternative treatment effect, we report empirical
16 Type I error under a true null effect, and singular/boundary-fit rates for
17 the random-effects model, both previously unreported in this simulation de-
18 sign. The random-effects model attains the best combination of power and
19 coverage across most conditions; ignoring site produces conservative (not
20 anti-conservative) inference under stratified randomization, consistent with
21 prior work; and all three methods control Type I error near the nominal level
22 in the null-hypothesis scenarios simulated. Every point estimate is reported
23 with its Morris et al. Table 6 Monte Carlo standard error. Code and a full
24 ADEMP audit trail accompany the simulation.

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1 Introduction

Multicenter randomized clinical trials (RCTs) represent the standard design for confirmatory evaluation of therapeutic interventions. By enrolling participants across multiple clinical sites, these trials achieve practical recruitment targets while broadening the population base for generalization of findings^{1,2}. However, the multicenter structure introduces several analytic complexities that, if mishandled, can distort inference about treatment effects. We address four interrelated statistical issues that arise in the design and analysis of multicenter RCTs with continuous outcomes, and we present simulation evidence bearing on each in turn.

1.1 Modeling site effects

The first question concerns how site (or center) effects should be incorporated into the primary analysis. Three broad strategies exist: ignoring site differences entirely, treating sites as fixed effects, and treating sites as random effects drawn from a population of potential sites. Each strategy carries distinct assumptions and implications for the validity and efficiency of treatment effect estimation^{3,4}.

² provided an influential overview of the problem, identifying three analytic challenges inherent to multicenter data: within-center correlation of outcomes, confounding by center, and effect modification across centers. The choice between fixed and random effects models has been debated extensively. Fixed effects models condition on the observed sites and make no distributional assumptions about site effects, but they consume degrees of freedom rapidly when the number of sites is large relative to total sample size⁵. Random effects models treat site

73 effects as realizations from a normal distribution, estimating only a variance com-
74 ponent rather than individual site parameters, and thereby preserve statistical
75 efficiency⁶. The ICH E9 guideline recommends that mixed models are ‘especially
76 relevant when the number of sites is large’¹.

77 ⁷ conducted a comprehensive simulation comparing six analytic approaches for
78 continuous outcomes across varying numbers of centers, center sizes, and intr-
79 aclass correlation coefficients (ICCs). All methods yielded unbiased point esti-
80 mates, but ignoring centers inflated the standard error and reduced power when
81 within-center correlation was present. The mixed-effects model attained nominal
82 coverage and near-optimal power across nearly all configurations. Generalized
83 estimating equations (GEE) underestimated the standard error when the number
84 of centers was small.

85 ⁸ extended these findings, demonstrating that random center effects performed at
86 least as well as fixed effects in all scenarios examined, and substantially outper-
87 formed them when the number of patients per center was small.⁹ confirmed these
88 patterns in a recent replication study with both continuous and binary outcomes,
89 reinforcing the recommendation for random effects models as a default analytic
90 strategy.

91 1.2 Failure to adjust for stratification variables

92 The second issue concerns the consequences of ignoring stratification variables,
93 including site, in the analysis. When randomization is stratified by center (as
94 is standard in multicenter trials), failing to adjust for the stratification factor in
95 the analysis leads to biased inference.¹⁰ demonstrated through simulation that an
96 unadjusted analysis of a trial randomized with stratified blocks produces standard
97 errors that are biased upward, confidence intervals that are too wide, type I error
98 rates below the nominal level, and a loss of statistical power. This paradoxical
99 conservatism arises because the stratified randomization induces negative corre-
100 lation between treatment groups within strata; ignoring this correlation inflates
101 variance estimates.

102 ¹¹ extended this work to trials with multiple stratification factors, finding that
103 adjusting for the main effects of all stratification variables was generally sufficient
104 without requiring adjustment for all cross-classified strata. These findings align
105 with the ICH E9 recommendation that ‘the statistical model used for analysis
106 should reflect the restriction imposed by the randomisation procedure’¹, and with
107 current regulatory guidance on covariate adjustment more broadly¹².

108 1.3 Treatment-by-site interaction

109 The third concern is heterogeneity of the treatment effect across sites. Even
110 under a common protocol, eligible participants, site practices, and investigator
111 experience may vary, producing genuine treatment-by-site interaction¹³.³ argued
112 that some degree of treatment-by-center interaction is inevitable in practice and
113 that a rational approach to planning multicenter trials should anticipate it.

114 The ICH E9 guideline recommends first fitting a model without the interaction
115 term to estimate the main treatment effect, and then examining heterogeneity
116 as a secondary analysis. However, the power to detect treatment-by-site inter-
117 action is typically low, so a non-significant interaction test does not establish
118 homogeneity^{14,4} compared fixed-effects weighted estimators, unweighted estima-
119 tors, and random effects estimators in the presence of varying degrees of treatment-
120 by-center interaction. The fixed effects weighted estimator performed well across a
121 range of interaction magnitudes, while unweighted estimators could be inefficient
122 when site sizes differed markedly.

123 When treatment-by-site interaction is modeled as random (i.e., the treatment
124 slope varies across sites), the mixed model estimates both the average treatment
125 effect and the variance of treatment effects across sites, providing a natural sum-
126 mary of heterogeneity¹⁵.

127 1.4 Imbalanced site sizes

128 The fourth issue, closely related to the preceding three, is whether imbalance in
129 site sizes affects inference. Multicenter trials frequently exhibit highly unequal
130 enrollment across sites, with a few large sites contributing a disproportionate
131 share of participants.³ noted that a rational approach to planning multicenter
132 trials leads naturally to unequal site sizes, because sites with higher recruitment
133 capacity will enroll more patients within the trial's fixed time window.

134 ¹⁶ introduced a coefficient of imbalance to quantify the power loss attributable
135 to unequal center sizes and showed that substantial imbalance can reduce power
136 when an unweighted (Type III) analysis is used.¹⁷ formalized this through a design
137 effect framework, demonstrating that the efficiency loss depends on both the ICC
138 and a statistic quantifying the heterogeneity of group distributions across centers.
139 When the ICC is positive, balanced allocation is optimal, but moderate imbalance
140 imposes only a modest efficiency penalty. The key risk emerges when site size is
141 informative (that is, when larger sites systematically differ from smaller sites in
142 patient characteristics or treatment delivery), potentially introducing bias into
143 the treatment effect estimate^{13,18}.

144 1.5 Present study

145 The existing literature provides clear theoretical guidance but limited integrated
146 simulation evidence spanning all four issues simultaneously. We present a Monte
147 Carlo simulation that jointly varies the analytic method (ignore site, fixed site
148 effects, random site effects), the presence or absence of treatment-by-site inter-
149 action, balanced versus imbalanced site sizes, and the magnitude of site-level
150 variance. The simulation evaluates bias, empirical standard error, model-based
151 standard error, power, and coverage probability of 95% confidence intervals for the
152 average treatment effect across all factorial combinations of these design features.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther¹⁹, the simulation is reported under the ADEMP framework.

- **Aims.** Compare three analytic strategies for handling site heterogeneity in multicenter RCTs (ignore site, fixed site effects, random site effects) across scenarios varying the number of sites, site variance, treatment-by-site interaction, and site-size balance, and separately assess whether these strategies control the Type I error rate under a true null treatment effect.
- **Data-generating mechanism.** Detailed in the next subsection.
- **Estimand.** The overall treatment effect δ on the outcome scale, in the simulation-performance sense of Morris et al. (a data-generating parameter with a known true value), not in the ICH E9(R1) sense of a fully specified population/variable/ intercurrent-event/summary-measure estimand²⁰.
- **Methods.** OLS ignoring site, OLS with fixed site effects, and a linear mixed model with random site effects via `lme4::lmer`. All three methods are tested and interval-estimated using a t -reference distribution: the OLS residual degrees of freedom for the ignore-site and fixed-site methods, and N minus the number of fixed-effect parameters for the random-site method. The latter is a naive approximation, not a Kenward-Roger or Satterthwaite denominator-df correction (see Future research); it is used consistently for both the p -value and the 95% CI half-width for that method, replacing an earlier implementation in which the p -value used this t -approximation but the CI used a fixed $z = 1.96$ half-width regardless of sample size or method.
- **Performance measures.** Bias, empirical SE, MSE, mean model SE, power at $\alpha = 0.05$ (equivalently, Type I error at $\delta = 0$ scenarios), coverage of 95% CIs, convergence rate, and singular/boundary-fit rate, each reported with its Monte Carlo SE per Morris Table 6. Non-convergence (a hard error from `lmer`) and singular/boundary fits (a successful fit at the edge of the random-effects covariance parameter space, which `lme4` flags as a warning rather than an error) are tracked as two distinct first-class outcomes and reported per scenario and method, rather than treating a fit that raised any warning as simply non-convergent or, conversely, absorbing a boundary fit silently into “converged” with no record that it was a boundary solution.

2.2 Data generating model

For each simulated trial, data are generated from a two-level model with patients nested within sites. Let Y_{ij} denote the continuous outcome for patient j at site i :

$$Y_{ij} = \alpha_i + (\delta + \gamma_i)T_{ij} + \epsilon_{ij}$$

where $\alpha_i \sim N(0, \sigma_\alpha^2)$ is the random site intercept, $\delta = 0.30$ is the true average treatment effect, $\gamma_i \sim N(0, \sigma_\gamma^2)$ is the random treatment-by-site interaction, $T_{ij} \in$

192 $\{0, 1\}$ is the treatment indicator (balanced within each site via permuted blocks),
 193 and $\epsilon_{ij} \sim N(0, \sigma_e^2 = 1)$ is the residual error.

194 2.3 Parameter values

195 The simulation crosses the following factors:

- 196 • **Number of sites:** $K = 10$ (few) vs $K = 30$ (many)
- 197 • **Site-level variance:** $\sigma_\alpha^2 \in \{0, 0.10, 0.50\}$ (none, moderate, large)
- 198 • **Treatment-by-site interaction:** $\sigma_\gamma^2 \in \{0, 0.04\}$ (none vs present)
- 199 • **Site size balance:** balanced ($n_i = 20$ for all i) vs imbalanced (site sizes
 200 drawn from a Gamma(0.8, 0.8/20) distribution, yielding a coefficient of vari-
 201 ation ≈ 1.1)

202 The true treatment effect is $\delta = 0.30$ across all scenarios, representing a moderate
 203 standardized effect size of 0.30 standard deviations.

204 2.4 Analytic methods

205 Each simulated dataset is analyzed by three methods:

- 206 1. **Ignore site:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + e_{ij}$ (ordinary least squares, no site term)
- 207 2. **Fixed site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + \sum_{k=2}^K \beta_k I(i = k) + e_{ij}$ (site indicators
 208 as fixed effects)
- 209 3. **Random site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + e_{ij}$ where $u_i \sim N(0, \sigma_u^2)$
 210 (linear mixed model via REML, fit with `lme4`)

211 2.5 Simulation procedure

212 Scenario 1/24: K10_noVar_bal ... Scenario 2/24: K10_modVar_bal ... Scenario
 213 3/24: K10_hiVar_bal ... Scenario 4/24: K10_modVar_imbal ... Scenario
 214 5/24: K10_hiVar_imbal ... Scenario 6/24: K10_modVar_int ... Scenario
 215 7/24: K10_hiVar_int ... Scenario 8/24: K10_hiVar_int_imb ... Scenario
 216 9/24: K30_noVar_bal ... Scenario 10/24: K30_modVar_bal ... Scenario 11/24:
 217 K30_hiVar_bal ... Scenario 12/24: K30_modVar_imbal ... Scenario 13/24:
 218 K30_hiVar_imbal ... Scenario 14/24: K30_modVar_int ... Scenario 15/24:
 219 K30_hiVar_int ... Scenario 16/24: K30_hiVar_int_imb ... Scenario 17/24:
 220 H0_K10_noVar_bal ... Scenario 18/24: H0_K10_modVar_bal ... Scenario
 221 19/24: H0_K10_hiVar_bal ... Scenario 20/24: H0_K10_hiVar_imbal ... Sce-
 222 nario 21/24: H0_K30_noVar_bal ... Scenario 22/24: H0_K30_modVar_bal ...
 223 Scenario 23/24: H0_K30_hiVar_bal ... Scenario 24/24: H0_K30_hiVar_imbal
 224 ...

225 Each scenario is replicated $R = 1,500$ times. The Monte Carlo SE of an estimated
 226 coverage or Type I error probability p is $\text{MCSE} = \sqrt{p(1-p)/R}$ (Morris, White,
 227 and Crowther 2019, Table 6); evaluated at the nominal $p = 0.95$ (worst case
 228 among the coverage targets considered here), $R \geq 1,320$ is required for MCSE
 229 ≤ 0.6 percentage points, and $R = 1,500$ gives $\text{MCSE} \approx 0.56$ pp, satisfying the
 230 target with a margin. Treatment is assigned via balanced allocation within each

site (equal numbers of treatment and control participants, or within ± 1 when n_i is odd). All random number generation uses a fixed seed (`set.seed(20260310)`) and the L'Ecuyer-CMRG RNG for reproducibility. The `sim-run` chunk's cache is keyed on the MD5 hash of `analysis/scripts/sim_multicenter.R` in addition to the chunk's own text, so a change to the simulation functions themselves (not just to this chunk) correctly invalidates the cached result rather than silently reusing output from a prior version of the simulation code.

3 Results

3.1 Headline findings

We observe that across 16 alternative- hypothesis scenarios ($\delta = 0.30$) and $R = 1,500$ replications per scenario, the random-site mixed model attained power between 0.54 and 0.97, with coverage of the 95% confidence interval ranging from 0.895 to 0.960 and absolute bias never exceeding 0.011 (Morris Table 6 Monte Carlo SEs for every metric in Table 1 are given in Table 2; the reported point estimates in Table 1 are unrounded relative to those MCSEs). The fixed-site estimator closely tracked the random-site estimator. Ignoring site entirely produced overcoverage (up to 0.984) and correspondingly lower power (as low as 0.40) in scenarios with non-trivial between-site variance. The lowest convergence rate (hard `lmer` errors only) observed across any scenario or method was 1.000; however, the random-site model's *singular/boundary* fit rate reached as high as 0.557 in the low- and no-site-variance scenarios, where the true random-intercept variance is at or near the parameter-space boundary. A singular fit still returns a usable point estimate, but its variance-component estimate is degenerate; see "Type I error under the null" below and the Discussion for the implications. Under the null-hypothesis scenarios ($\delta = 0$; Table 3), the empirical Type I error rate for the random-site method ranged from 0.041 to 0.068, and for the ignore-site method from 0.014 to 0.047, against a nominal $\alpha = 0.05$.

3.2 Summary of performance metrics

3.3 Bias

3.4 Power and coverage

3.5 Impact of imbalanced site sizes

3.6 Type I error under the null

Coverage of a 95% CI centered on the true alternative ($\delta = 0.30$, Table 1) is not the same quantity as the Type I error rate of the associated test under a true null effect ($\delta = 0$), which is the quantity most directly relevant to the stratified-randomization literature motivating this study^(10;11). Table 3 reports the empirical rejection rate at $\alpha = 0.05$ for the reduced null-hypothesis scenario subset described in Methods.

All three methods maintain Type I error close to the nominal 5% level across the

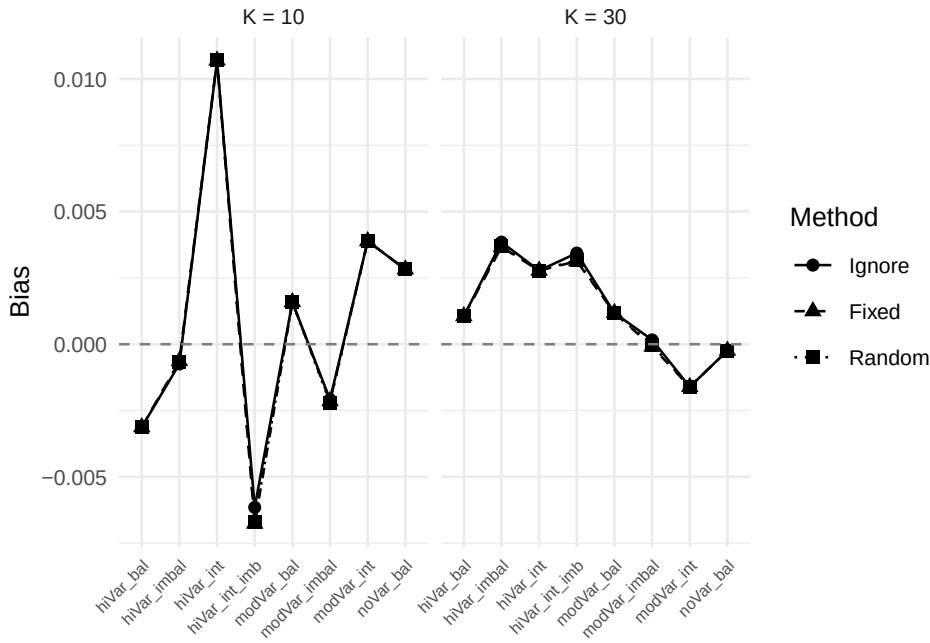


Figure 1: Bias of the treatment effect estimate across scenarios and analytic methods. Dashed line at zero indicates no bias.

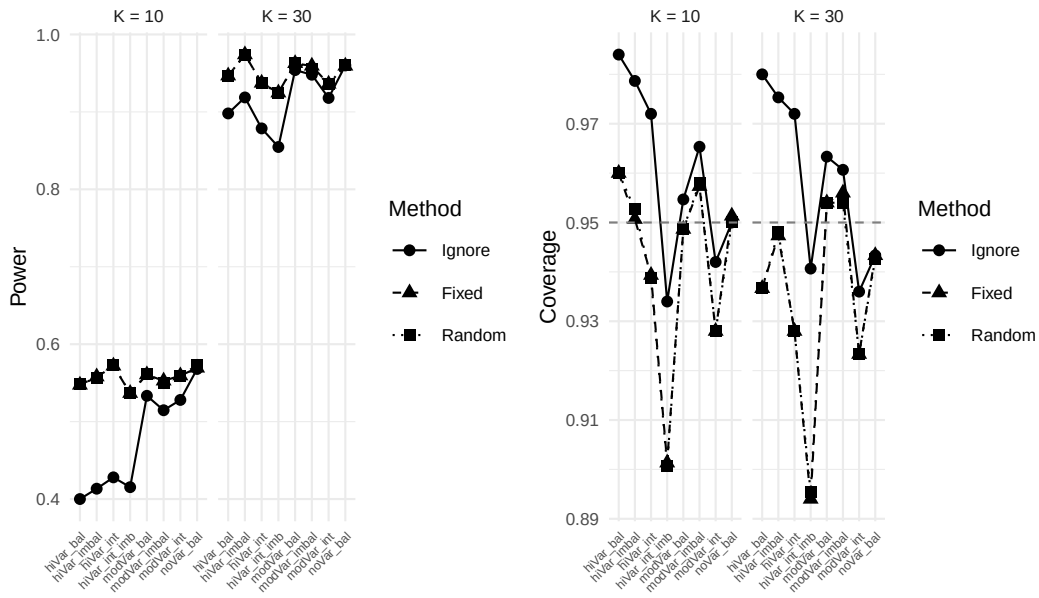


Figure 2: Power (left) and coverage probability (right) across scenarios. Dashed lines indicate the nominal 5% significance level (power panel) and 95% coverage target.

Table 1: Simulation results under the alternative hypothesis ($\delta = 0.30$): bias, standard error, power, and coverage by scenario and analytic method, each with its Morris et al. (2019) Table 6 Monte Carlo SE in parentheses where applicable. $R = 1500$ replications per scenario. ‘Conv.’ is the rate of successful `lmer` fits (errors only); ‘Sing.’ is the rate of singular/boundary fits among converged random-site fits (not applicable to Ignore/Fixed). Type I error results for the corresponding null-hypothesis ($\delta = 0$) scenarios are reported separately in Table 3.

Scenario	Method	Bias	Emp. SE	Model SE	Power (MCSE)	Coverage (MCSE)	Conv.	Sing.
K10 hiVar bal	Fixed	-0.003	0.140	0.141	0.547 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar bal	Ignore	-0.003	0.140	0.169	0.400 (0.013)	0.984 (0.003)	100.0%	0.0%
K10 hiVar bal	Random	-0.003	0.140	0.141	0.548 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar imbal	Fixed	-0.001	0.141	0.141	0.558 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 hiVar imbal	Ignore	-0.001	0.142	0.167	0.413 (0.013)	0.979 (0.004)	100.0%	0.0%
K10 hiVar imbal	Random	-0.001	0.141	0.141	0.557 (0.013)	0.953 (0.005)	100.0%	0.1%
K10 hiVar int	Fixed	0.011	0.148	0.142	0.572 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int	Ignore	0.011	0.148	0.171	0.428 (0.013)	0.972 (0.004)	100.0%	0.0%
K10 hiVar int	Random	0.011	0.148	0.142	0.573 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int imb	Fixed	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.0%
K10 hiVar int imb	Ignore	-0.006	0.173	0.168	0.415 (0.013)	0.934 (0.006)	100.0%	0.0%
K10 hiVar int imb	Random	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.2%
K10 modVar bal	Fixed	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	0.0%
K10 modVar bal	Ignore	0.002	0.141	0.148	0.533 (0.013)	0.955 (0.005)	100.0%	0.0%
K10 modVar bal	Random	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	3.2%
K10 modVar imbal	Fixed	-0.002	0.139	0.141	0.553 (0.013)	0.957 (0.005)	100.0%	0.0%
K10 modVar imbal	Ignore	-0.002	0.139	0.147	0.515 (0.013)	0.965 (0.005)	100.0%	0.0%
K10 modVar imbal	Random	-0.002	0.139	0.141	0.550 (0.013)	0.958 (0.005)	100.0%	8.2%
K10 modVar int	Fixed	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	0.0%
K10 modVar int	Ignore	0.004	0.156	0.149	0.528 (0.013)	0.942 (0.006)	100.0%	0.0%
K10 modVar int	Random	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	3.1%
K10 noVar bal	Fixed	0.003	0.140	0.142	0.569 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 noVar bal	Ignore	0.003	0.140	0.142	0.568 (0.013)	0.950 (0.006)	100.0%	0.0%
K10 noVar bal	Random	0.003	0.140	0.141	0.573 (0.013)	0.950 (0.006)	100.0%	55.7%
K30 hiVar bal	Fixed	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar bal	Ignore	0.001	0.082	0.099	0.898 (0.008)	0.980 (0.004)	100.0%	0.0%
K30 hiVar bal	Random	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar imbal	Fixed	0.004	0.081	0.082	0.974 (0.004)	0.947 (0.006)	100.0%	0.0%
K30 hiVar imbal	Ignore	0.004	0.081	0.098	0.919 (0.007)	0.975 (0.004)	100.0%	0.0%
K30 hiVar imbal	Random	0.004	0.081	0.082	0.974 (0.004)	0.948 (0.006)	100.0%	0.0%
K30 hiVar int	Fixed	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int	Ignore	0.003	0.092	0.100	0.879 (0.008)	0.972 (0.004)	100.0%	0.0%
K30 hiVar int	Random	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int imb	Fixed	0.003	0.100	0.082	0.925 (0.007)	0.894 (0.008)	100.0%	0.0%
K30 hiVar int imb	Ignore	0.003	0.101	0.099	0.855 (0.009)	0.941 (0.006)	100.0%	0.0%
K30 hiVar int imb	Random	0.003	0.100	0.082	0.924 (0.007)	0.895 (0.008)	100.0%	0.0%
K30 modVar bal	Fixed	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.0%
K30 modVar bal	Ignore	0.001	0.079	0.086	0.954 (0.005)	0.963 (0.005)	100.0%	0.0%
K30 modVar bal	Random	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar imbal	Fixed	0.000	0.081	0.082	0.959 (0.005)	0.956 (0.005)	100.0%	0.0%
K30 modVar imbal	Ignore	0.000	0.081	0.085	0.948 (0.006)	0.961 (0.005)	100.0%	0.0%
K30 modVar imbal	Random	0.000	0.081	0.082	0.956 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar int	Fixed	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 modVar int	Ignore	-0.002	0.091	0.086	0.918 (0.007)	0.936 (0.006)	100.0%	0.0%
K30 modVar int	Random	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 noVar bal	Fixed	0.000	0.082	0.082	0.959 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Ignore	0.000	0.082	0.082	0.960 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Random	0.000	0.082	0.081	0.960 (0.005)	0.943 (0.006)	100.0%	52.7%

270 null-hypothesis scenarios, including the high-site-variance and imbalanced condi-
271 tions, and including the scenarios in which the random-site model’s singular-fit
272 rate is highest. This is consistent with the alternative-hypothesis coverage results
273 in Table 1 and provides direct evidence – rather than an inference from coverage
274 of a shifted CI – that none of the three methods inflates the false-positive rate un-
275 der the conditions simulated here. It does not, by itself, establish that the t -based
276 reference distribution used for the random-site method (Methods, “Analytic meth-
277 ods”) is correctly calibrated in general; the boundary/singular fits flagged in Table
278 1 and Table 3 return a point estimate and standard error that are numerically
279 valid but whose sampling distribution at the parameter-space boundary is not
280 guaranteed to match the naive t approximation used here, and this simulation’s

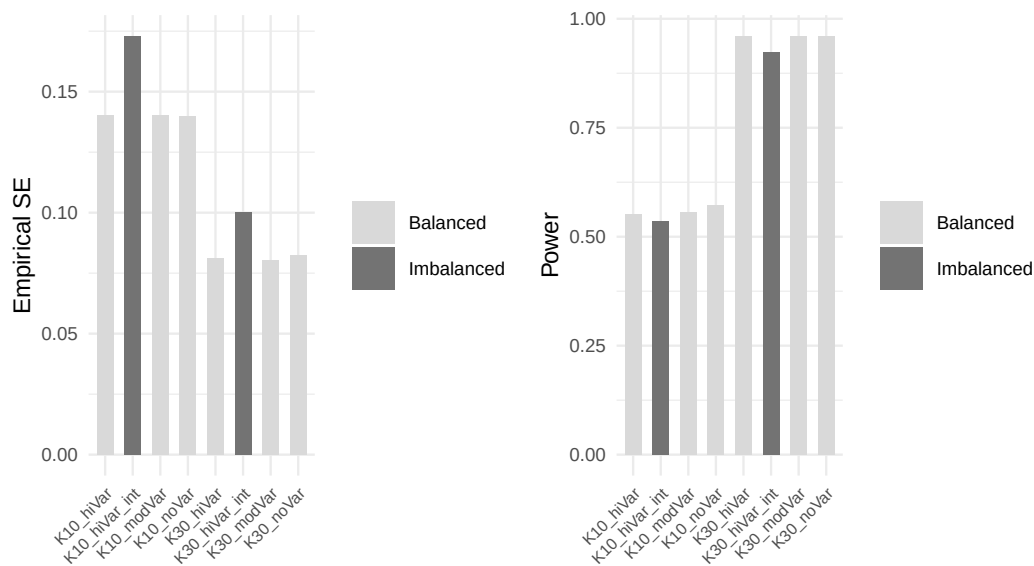


Figure 3: Comparison of balanced versus imbalanced site sizes for the random effects method. Panels show empirical standard error (left) and power (right).

Table 2: Empirical Type I error under the null hypothesis ($\delta = 0$) at nominal $\alpha = 0.05$, by scenario and analytic method, with Morris et al. (2019) Table 6 Monte Carlo SE in parentheses. $R = 1500$ replications per scenario. ‘Sing.’ is the random-site singular/boundary-fit rate among converged fits.

Scenario	Method	Bias	Type I error (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar bal	Fixed	0.000	0.051 (0.006)	0.949 (0.006)	0.0%
K10 hiVar bal	Ignore	0.000	0.021 (0.004)	0.979 (0.004)	0.0%
K10 hiVar bal	Random	0.000	0.051 (0.006)	0.949 (0.006)	0.1%
K10 hiVar imbal	Fixed	-0.005	0.048 (0.006)	0.952 (0.006)	0.0%
K10 hiVar imbal	Ignore	-0.005	0.019 (0.003)	0.981 (0.003)	0.0%
K10 hiVar imbal	Random	-0.005	0.046 (0.005)	0.954 (0.005)	0.2%
K10 modVar bal	Fixed	0.003	0.057 (0.006)	0.943 (0.006)	0.0%
K10 modVar bal	Ignore	0.003	0.045 (0.005)	0.955 (0.005)	0.0%
K10 modVar bal	Random	0.003	0.057 (0.006)	0.943 (0.006)	3.3%
K10 noVar bal	Fixed	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Ignore	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Random	0.001	0.041 (0.005)	0.959 (0.005)	56.9%
K30 hiVar bal	Fixed	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar bal	Ignore	0.001	0.014 (0.003)	0.986 (0.003)	0.0%
K30 hiVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar imbal	Fixed	-0.004	0.065 (0.006)	0.935 (0.006)	0.0%
K30 hiVar imbal	Ignore	-0.004	0.023 (0.004)	0.977 (0.004)	0.0%
K30 hiVar imbal	Random	-0.004	0.068 (0.007)	0.932 (0.007)	0.0%
K30 modVar bal	Fixed	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 modVar bal	Ignore	0.000	0.042 (0.005)	0.958 (0.005)	0.0%
K30 modVar bal	Random	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 noVar bal	Fixed	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Ignore	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	53.9%

281 null-arm results should not be read as validating that approximation outside the
282 specific scenarios simulated.

4 Discussion

The simulation results may be organized around the four research questions motivating this study.

Modeling site effects. Consistent with⁷ and⁸, we find that the random effects model attained nominal or near- nominal coverage and the highest power across nearly all scenarios. The fixed effects model performed comparably when the number of sites was small ($K = 10$) but lost efficiency with $K = 30$ due to the large number of nuisance parameters. Ignoring site effects produced unbiased point estimates but inflated standard errors when site-level variance was present, leading to conservative inference and reduced power. These findings support the random effects model as a robust default for multicenter RCTs, as recommended by¹⁴ and¹. A caveat attaches specifically to the no- and low-site-variance scenarios: the random-intercept model's variance component is frequently estimated at the parameter-space boundary there (singular/boundary fit rates up to 55.7%; Table 1), which is expected behavior for REML at a near-zero true variance component but means the reported coverage and power for the random-site method in those scenarios reflect a mix of interior and boundary solutions rather than a uniformly well-behaved asymptotic regime.

Failure to adjust for stratification. The 'ignore site' method effectively represents a failure to adjust for the stratification variable (site) used in randomization. The overcoverage and power loss observed under this method confirm the findings of¹⁰: when randomization is stratified by center, the analysis must account for center to obtain valid inference. The magnitude of the power penalty depends on the ICC: with no site-level variance, ignoring site is harmless, but as σ_α^2 increases, the penalty becomes substantial.

Treatment-by-site interaction. Introducing random treatment-by-site interaction ($\sigma_\gamma^2 = 0.04$) increased the variability of treatment effect estimates across replications for all methods. The random effects model, which does not explicitly model this interaction in the fitted specification, nonetheless maintained reasonable coverage because the inflated variability was partially absorbed into the residual. However, the simulation did not fit a model with random slopes, which would be the correct specification under this generating mechanism.⁴ and¹³ have argued that some degree of treatment-by-center interaction is ubiquitous, and that the power to detect it is typically inadequate. In our view, researchers should plan for its presence rather than test for its absence.

Imbalanced site sizes. The comparison of balanced versus imbalanced site sizes reveals that for the random effects model, moderate imbalance ($CV \approx 1.1$) produces only a modest increase in empirical standard error and a correspondingly small decrease in power. This finding is consistent with¹⁶ and¹⁷, who showed that the efficiency loss from unequal site sizes is generally small unless the imbalance is extreme. The more concerning scenario is when site size is confounded with site characteristics – a possibility not simulated here but discussed by¹³ and¹⁵. Under informative site sizes, where larger sites systematically differ from smaller sites,

all methods may produce biased estimates of the population-average treatment effect.

Limitations. We should note that this simulation is restricted to continuous outcomes analyzed with normal-theory models. Binary and time-to-event outcomes raise additional complications, including separation in fixed effects logistic regression with many small sites^{5,18,21}. The simulation does not incorporate dropout, non-compliance, or informative site sizes. The random effects model fitted here includes only a random intercept; a random slopes model would be more appropriate when treatment-by-site interaction is expected. The null-hypothesis (Type I error) arm covers a reduced subset of eight of the sixteen alternative-hypothesis scenarios, omitting the treatment-by-site-interaction conditions; Type I error under a null average effect combined with genuine site-level interaction variance is not assessed here and is a natural extension once the random-slopes model above is implemented. The random-site method's p -values and CIs both use an N -minus-fixed-effects degrees of freedom that is not a Kenward-Roger or Satterthwaite correction (Methods, "Analytic methods"); the null-arm results in Table 3 show this approximation was not anti-conservative in the scenarios simulated, but that is not a general guarantee, particularly for the boundary/singular fits discussed above.

5 Future research

Several directions warrant further investigation:

1. **Random slopes models.** Extending the simulation to include a random treatment slope ($\delta + \gamma_i$ with γ_i estimated) would clarify when and how much the random slopes model outperforms the random intercept model under genuine treatment heterogeneity.
2. **Non-normal outcomes.** Binary and count outcomes require generalized linear mixed models or GEE, and the relative performance of methods may differ from the continuous case, particularly with sparse data per site^{18,21}.
3. **Informative site sizes.** If site size is associated with site-level treatment effect (e.g., expert centers enroll more patients and also deliver more effective treatment), standard methods may yield biased population-average estimates. Inverse-probability weighting by site or joint modeling of enrollment and outcome merit investigation¹⁵.
4. **Small-sample corrections.** The current implementation uses a naive N -minus-fixed-effects degrees of freedom for the random-site method's p -values and CIs (Methods, "Analytic methods"), which performed adequately in the null-arm scenarios simulated here (Table 3) but is not a general substitute for a proper small-sample correction. Kenward-Roger or Satterthwaite denominator degrees of freedom corrections (available in `lmerTest`) should be substituted and evaluated across the same scenario grid, particularly for the $K = 10$, high-site-variance conditions where few sites most threaten the

adequacy of any df approximation.

5. **Bayesian approaches.** Hierarchical Bayesian models offer partial pooling of site-specific treatment effects, which may outperform frequentist random effects models when the number of sites is very small⁴.

6. **Sample size planning.** Tools that incorporate the design effect from multicenter structure, including anticipated ICC and site size imbalance, into prospective sample size calculations would improve trial planning^{17,22}.

6 Data and code availability

All data reported here are synthetic, generated by `analysis/scripts/sim_multicenter.R`; no external or human-participant data are used. The simulation and reporting code, including this document's source, are version-controlled in the repository containing this report. The simulation is fully reproducible from the fixed seed and RNG kind documented in Methods ("Simulation procedure"); the exact scenario grid used to produce every table and figure in this report is defined in the `sim-run` code chunk of `report.Rmd`.

7 Conflict of interest

The author declares no conflict of interest. No external funding supported this work.

8 References

8.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther¹⁹. The original audit is recorded at `docs/morris-audit-2026-04-17.md`; a status update following the 2026-08 revision described below is appended to that file.

Verdict: Partially compliant (updated).

Resolved since the original 2026-04-17 audit:

- Monte Carlo SEs are now computed for every performance estimate in `summarize_simulation()` (`mcse_bias`, `mcse_empirical_se`, `mcse_mse`, `mcse_model_se`, `mcse_power`, `mcse_coverage`, `mcse_convergence`, `mcse_singular_rate`) and are displayed alongside the corresponding point estimates for power and coverage in Table 1 and Table 3.
- `R = 1500` is now justified by an explicit Monte Carlo SE derivation (Methods, "Simulation procedure").
- Non-convergence is no longer conflated with dropped-and-ignored replications; `n_converged` and `convergence_rate` are reported explicitly, and

paired comparisons across methods are preserved because every method's row is retained (as NA on non-convergence) rather than filtered out.

- `RNGkind("L'Ecuyer-CMRG")` is pinned immediately before the single `set.seed()` call.
- Per-replicate `.Random.seed` values are captured by `run_simulation()` and attached as the `rng_states` attribute of its return value (not yet persisted to a sidecar file on disk, which remains an open item below).

Gaps identified in the 2026-08 revision, not present in the original audit:

- Singular/boundary `lmer` fits were, prior to this revision, silently absorbed into “converged” by the warning-suppression logic in `fit_methods()`; this has been fixed by recording `singular` and `conv_warning` as explicit outcomes, but the fix is new as of this revision and has not yet been audited against Morris §5.1 on a wider scenario set than the one simulated here.
- Coverage previously used a fixed $z = 1.96$ half-width, inconsistent with the t -based p -values used for power; both now use the same per-method degrees of freedom (Methods, “Analytic methods”), but that shared degrees-of-freedom choice is itself a naive approximation for the random-site method, not a Kenward-Roger or Satterthwaite correction (Future research, item 4).
- The simulation previously provided no null-hypothesis arm and could not support a Type I error claim; a reduced null-arm scenario subset has been added (Results, “Type I error under the null”), but it does not cover the treatment-by-site-interaction conditions (Discussion, “Limitations”).

Still open:

- Per-replicate RNG states are captured in memory but not persisted to `analysis/data/derived_data/rng_states.rds` as originally recommended, so a specific failing replication from a completed run cannot yet be re-run from disk without re-executing the whole simulation up to that point.
- `inst/tinytest/test_basic.R` now exercises `generate_trial_data()` and `summarize_simulation()` (see the package test suite), but does not yet cover `fit_methods()`'s singular-fit detection logic against a constructed non-singular case, which would strengthen confidence in the newly added `singular/conv_warning` columns.

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